

MetalSynth-Pelvis: Conditioned synthesis of osteosynthesis implants and local metal artifacts in pelvic CT scan volumes using Multi-Window Image-Domain Diffusion

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1. Problem Statement

Automatic pelvic computed tomography (CT) analysis and volumetric segmentation depend on sufficiently annotated data. Metal artifacts can obscure anatomical boundaries near implants, motivating evaluation of segmentation robustness in this region. CTPelvic1K (Liu et al., 2021a) includes a CLINIC-metal subset whose limited bone annotation constrains supervised evaluation. Its name does not establish that every case contains pelvic osteosynthesis, nor that other subsets are free of metal or foreign objects. The local audit will therefore determine implant types, eligible training anatomy and annotated evaluation cases through three-dimensional and multiplanar review. The magnitude of peri-implant segmentation degradation must be measured in this study rather than assumed from the dataset description.

While generative models based on diffusion processes excel at synthesizing visually plausible medical anomalies, they structurally fail when applied to metallic implants. State-of-the-art diffusion-based lesion synthesis models fall into two families, and neither represents the intensity changes that the synthesized object induces outside itself. Mask-bounded inpainting methods condition generation on a lesion mask: CLAIM (Ramzan et al., 2026) recomposes the background from the original image at every denoising step, DiffTumor (Chen et al., 2024) models nothing outside the mask, and LGEsSynthNet (Jacob et al., 2026) is formulated as inpainting, although its latent compression still perturbs the whole slice incidentally rather than by design. Globally conditioned methods such as DiffBoost (Zhang et al., 2025b) instead synthesize the entire slice from noise, using the mask only as an edge constraint. None of these works states an assumption of tissue non-rigidity; what they share is that no mechanism, and no training signal, accounts for object-induced intensity changes beyond the mask. This is physically incompatible with metal implants, which are rigidly geometric and produce severe attenuation profiles (beam hardening, photon starvation, and high-amplitude streaking) that propagate beyond the implant's boundaries and degrade the signal of the surrounding tissue; reduction networks describe them as present globally in the reconstructed image (Li et al., 2024). De Man et al. (1999) attribute such streaks to beam hardening, scatter, noise and the exponential edge-gradient effect, and show them radiating from the metal along the directions of highest attenuation; their simulation, however, uses an iron rod and dental amalgam in a two-dimensional fan-beam geometry, excludes non-linear partial volume, and reports no spatial extent, so the size of the affected peri-implant band remains unmeasured in the literature.

Physics-based metal artifact simulation provides a reference for appearance generation. This thesis adopts the hybrid simulation protocol of Peters et al. (2025), implemented with CatSim within XCIST (De Man et al., 2007; Wu et al., 2022), as its physical comparison protocol. Their validation compares simulations against a physical phantom scanned on a commercial scanner; it does not cover the hybrid step (clinical images as input, with an empirical frequency-boosting compensation), the vendor-neutral geometry used to generate their data, or pelvic osteosynthesis in three dimensions, and the paper does not state whether patient and metal are projected jointly, which this work verifies in the simulator code before reproduction. Their training data use randomized metal placement, whereas their scoring benchmark uses expert placement. The unresolved placement problem is therefore scalable, anatomically constrained placement, rather than scalable metal insertion itself.

Even if automated placement were achieved, mere anatomical containment is insufficient; a valid simulation must reflect true surgical admissibility. Traditional deterministic surgical planning pipelines (Liu et al., 2025) attempt to optimize a single "perfect" trajectory. Yet real-world clinical data reveal a different reality. Grading iliosacral screws on a four-level cortical-breach scale, Zwingmann et al. (2009) report two distinct postoperative-CT distributions: grade 0 in 69% of navigated placements (grades 1–3: 15%, 8%, 8%) against grade 0 in 40% of conventional placements (37%, 11.5%, 11.5%). That scale reaches iliosacral fixation through Smith et al. (2006), who apply it there, but it originates in the pedicle-screw literature: Gertzbein and Robbins (1990) tabulate canal encroachment from the medial pedicle border in six 2 mm bins rather than four numbered grades, anchor the 2 mm cut to the epidural space, extrapolate the 4 mm cut from a single myelographic CT, and state no justification for the 6 and 8 mm cuts. The four-grade form (0, <2, 2–4, >4 mm) appears literally in Mirza et al. (2003), who declare the thresholds taken from prior studies, measured perforation in cadaveric thoracic spines rather than on CT, and caution that these "safe zone" thresholds do not apply to the thoracic spine and are likely different for different directions of screw perforation. The 2 mm bin width is therefore adopted here as a geometric convention, as with the 10 mm corridor criterion below, and not as a validated clinical threshold; its transfer from the spinal canal to the iliosacral corridor is an explicit assumption of this work. These are two surgical-technique populations that differ significantly, not one aggregate range; this work therefore targets the ordinal distributions themselves rather than a single imperfection rate. The distinction is not merely presentational. In a meta-analysis by the same first author, Zwingmann et al. (2013) pool malposition rates of 2.6% for conventional placement against 0.1% under CT navigation, and the 2009 cohort enters that pool with zero events in both arms, because the included series record malposition only where a revision followed; the authors themselves note that no comparable grading systems are given across studies. Aggregate malposition rates and ordinal cortical-breach grades therefore measure different constructs, and only the latter is a usable target for a placement sampler. Sacral level also warrants separate treatment on anatomical grounds. Using multiplanar CT measurements, Gardner et al. (2010) report mean minimum iliosacral cross-sectional areas at S1 and S2 of 346 and 109.3 mm² in normal sacra, compared with 222 and 220.1 mm² in dysmorphic sacra. The approximately equal areas in the latter group support measuring each level on the individual anatomy rather than assuming a universal corridor-size hierarchy. These measurements concern an uninjured side and ideal central trajectories; they do not describe clinical malposition frequencies. In a retrospective clinical series, van den Bosch et al. (2002) reported postoperative neurologic complaints in 6 of 31 patients with the lower screw in S2, versus 1 of 49 with both screws in S1 ($p = 0.01$). These are patient-level outcomes by screw configuration, not screw-malposition rates by level; the authors also identify a learning-curve bias associated with the change in placement practice. This clinical association supports retaining S1/S2 as a sampler variable, but supplies neither a causal level effect nor an ordinal cortical-breach prior for S2. Cortical breach by level need not follow the same ordering as patient-level neurologic outcome: on postoperative CT, Herman et al. (2017) report breach in 36.5% of S1 screws against 14.8% at S2 ($p = 0.035$), with 32% of screws breaching overall and no screw lying entirely outside bone. The two findings are measured on different quantities and are not in conflict; the latter bounds the range of malpositions the sampler must be able to produce. The ordinal benchmark from Zwingmann et al. (2009) applies to S1: the sacral level is explicit for the conventional arm and not reported for the navigated arm, where it is assumed to be S1, and grades are counted per screw, some patients having received more than one; placements in the second osseous corridor below S1 are evaluated descriptively for geometry and breach grades without claiming clinical distribution matching. Its grades were assigned on postoperative CT by one independent radiologist, with no slice thickness, reconstruction interval or inter-observer agreement reported, and the text does not state to which grade a perforation of exactly 2 or 4 mm belongs. Detection itself may depend on image resolution: Tejwani et al. (2014) describe

more foraminal penetration on 2.5 mm than on 5.0 mm sections, although at $P = .3$. Matching against this benchmark is therefore read as agreement between grade distributions measured at an unknown reference resolution, not as voxel-level equivalence, and the grade boundaries used here are a declared convention. The premise that no canonical pose exists is not only inferred from surgical error rates; it is supported anatomically. Measuring the orientation of the bony surfaces that bound the corridor across seventeen cadaveric pelves, Ziran et al. (2007) report inter-specimen coefficients of variation of 7–25% for most surfaces, rising to 43%, 47–52% and, for the orientation of the superior S1 ala in the sacral frontal plane, 97–140%. A corridor that varies this much between individuals cannot be served by one fixed trajectory. Training a diagnostic network solely on "perfect" deterministic trajectories will inevitably cause it to fail when encountering the true variance of clinical errors. Therefore, a generative sampler must not optimize for perfection, but instead sample a declared distribution of poses over the corridor measured on each volume, bounded by pelvic density representations (Arand et al., 2019) and by osseous-corridor viability. Note the direction of that statement: the sampler is not fitted to reproduce the clinical grade proportions. Fitting it to them and then scoring it against them would compare the model with the very data used to build it, and would leave the placement metric with nothing to validate. The clinical distributions are therefore held out as a target of comparison, not used as a calibration objective. For the latter, McLaren et al. (2021) supply the operational element that a qualitative safe-zone description (Ramadanov and Zabler, 2025) does not: a reproducible corridor-diameter procedure over 433 pelves, yielding transiliosacral angular tolerances of $1.53 \pm 0.57^\circ$ at S1 and $1.02 \pm 0.33^\circ$ at S2. Their ≥ 10 mm corridor criterion is not an empirically validated safety threshold and is not treated as one here: Kaiser et al. (2014), from whom it is taken, describe it as a chosen conservative value and justify it dimensionally, as clearance of 1–2 mm around a 6.3–8 mm screw. It is therefore used as a geometric clearance convention rather than as an inherited clinical standard. Kaiser et al. (2014) additionally supply what a corridor diameter alone does not: a reference frame computable on a CT volume, obtained by reformatting along the sacral axis perpendicular to the superior end plate of S1, with corridor angulation measured coronally against the line joining the iliac crests and axially against the line joining the posterior iliac spines, and a usable-length rule requiring at least 5 mm of clearance to the cortex on either side. The pose distribution sampled within that frame remains an explicit construction of this work.

Recognized Gap. Current attempts to bridge the metal-artifact domain shift are bottlenecked by three mutually exclusive paradigms: physical simulators do not supply scalable anatomically constrained placement, biological diffusion models cannot generate rigid global streaks, and deterministic surgical planners ignore the statistical reality of clinical malposition. To the best of the author’s knowledge, there is no diffusion-based method that overcomes these barriers by combining: **(C1)** Non-biological rigid implant geometries, which avoid the threshold-dependent size distortion of fixed-HU metal segmentation (over-coverage in simulated slices, Xie et al., 2024; fragmentation of pelvic screws in the local audit); **(C2)** A surgically constrained placement sampler that accurately emulates the population distribution of clinical malpositions; and **(C3)** The transfer of multi-window HU encoding from artifact *removal*, where it originates, to artifact *synthesis*, coupled with a generation band (B_δ) extended beyond the implant mask so that streaking and beam hardening are modelled in a bounded region outside the metal itself. The multi-window framework is not claimed as novel here: it is established in metal-artifact-reduction work, where published uses apply the windows to per-window reconstruction stages or to the training loss rather than to the encoding of the input (Wang et al., 2025; Li et al., 2024), and Wang et al. (2025) attribute it in turn to earlier multiple-window learning. What is claimed is its use for generation, together with B_δ , for which no published precedent quantifying a *generation band outside the implant mask* was found. Peri-implant artifact extent itself has been measured: Radzi et al. (2014) report CT artifact distances of “2.0, 2.6, 1.6 and 2.0 mm” across titanium, stainless-steel and cannulated screws, and Cassanego et al. (2026) report 3.1–4.2 mm by the same style of measurement. Neither supplies a calibration for B_δ . Both measure the “perpendicular distance from the central screw axis to the boundary of the artifact”, that is from the screw *axis* rather than from its surface, to the edge of a thresholded surface whose threshold is not published, on 3.5–4.0 mm screws in single-specimen ankle and dental settings; they do not report the quantity in Hounsfield units, and their two ranges are not mutually comparable for lack of a common reference point. The band width used here is therefore a declared construction, not an inherited measurement. This is the gap this work aims to fill.

Filling it rests on one assumption that the present literature neither grants nor refutes: that the *appearance* of a metal artifact can be generated in the image domain, without access to projection data. The mechanisms involved are non-linear and are manipulated in the sinogram (De Man et al., 1999), and dual-domain reduction networks report a penalty for operating on images alone: Lin et al. (2019) obtain 31.45 dB PSNR from their image-domain variant against 33.51 dB for the full dual-domain network. That image-domain variant is fed a reconstruction of an already-interpolated sinogram, but the penalty does not rest on this detail: Li et al. (2024), whose image-only branch starts from the metal-corrupted reconstruction, also report it below their multi-domain network and state that working in a single domain cannot effectively recover the true tissues. One qualification keeps these results from settling the question: both address artifact *removal*, which Lin et al. (2019) describe as ill-posed because information has been destroyed, whereas synthesis carries no such recovery requirement. This work therefore treats image-domain generability not as a premise but as part of what its distributional physical-coherence evaluation puts to the test, with the physics-based protocol of Peters et al. (2025) supplying the mechanism-faithful reference against which the generated appearance is judged.

2. Research Question

What improvement does a surgical admissibility-constrained (bone density and cortical containment) 3D placement sampler, a multi-window image-domain diffusion model conditioned on the rigid implant geometries and an extended generation band (B_δ), have in terms of distributional physical-coherence of the synthesized pelvic osteosynthesis implants and their local artifacts, when compared with both naive copy-paste insertion and the adopted physics-based simulation protocol of Peters et al. (2025)? Downstream segmentation impact is deliberately excluded from this study and is stated as out of scope below.

3. Hypothesis

The synthetic generation of medical anomalies and metallic implants using a rigid geometry and extended spatial boundaries, HU-conditioned diffusion model and whose placement is surgically enforced via a density-based placement sampler is more physically-coherent than unconstrained generative baseline methods and on the same footing as the adopted explicit physics-based simulation protocol (Peters et al., 2025). Placement coherence is quantified by Surgical Admissibility of Placement (SAP), the single metric introduced by this work, which emulates the measured ordinal distribution of real-world implant placement. Appearance coherence is quantified with the metrics of the adopted protocol under their published names, not with newly coined ones.

4. Objectives

General Objective. Design, implement and evaluate a physics-and-anatomy-aware image-domain diffusion pipeline with rigid surgical constraints and extended HU-based conditioning, capable of generating physically coherent metallic implants intended, in future work, to improve the robustness of downstream segmentation networks.

- 1. Validation of a multi-window representation (Compulsory Go/No-Go):** Ensure HU reconstruction error (HU to multi-window to VAE to HU again) within the bone tissue, delimited by the same 150 HU bone threshold used by the adopted evaluation protocol (Peters et al., 2025; Haneda et al., 2025), below a 25 HU threshold (Mean Absolute Error (MAE)). No published pass/fail threshold on CT-number accuracy was found, so this criterion is not taken from a source: it is anchored on the error level that current metal-artifact methods reach on the same scale. The NMAR algorithm that calibrates the community benchmark reports an RMSE of 20.2 HU, a sinogram-domain diffusion model evaluated in image space 12.3 HU (Karageorgos et al., 2024), and a latent diffusion model 12.74 HU (Yun et al., 2026). Because those values are RMSE while this criterion is MAE, and RMSE is never below MAE for the same residuals, the comparison fixes the order of magnitude rather than an equivalence; this is declared rather than resolved. A concurrent preprint reports that frozen video autoencoders transfer to CT without medical fine-tuning (Chen et al., 2026); it reports PSNR, SSIM and MSE without declared units and no HU error, evaluates no bone or metal region, uses no pelvic or metal-bearing volumes, and declares intensity clipping to $[-1000, 1000]$ HU for its generation experiments, so it does not address the dense-bone and metal range this criterion targets. In volumes with metal, the same round-trip error is additionally reported inside the band B_δ around the metal, descriptively and outside the Go/No-Go: that band holds the low-amplitude streaks the synthesizer must render, and an error restricted to bone does not show whether the autoencoder preserves them, since the first stage of latent diffusion is designed to remove high-frequency detail (Rombach et al., 2022). The gate rule is fixed before any run. It is evaluated on 34 held-out patients (patient-level split stratified by dataset and metal presence) for the pretrained autoencoder and for a decoder-adapted variant with the encoder frozen, each under three three-channel encodings: the windows reported by Wang et al. (2025) ($[-1000, 2000]$, $[-320, 480]$ and $[-160, 240]$ HU), the same with the wide-window ceiling raised to 20000 HU, and the wide window replaced by an arcsinh compression over $[-1000, 20000]$ HU. HU are read back without ground truth, from the narrowest non-saturated channel. A combination passes when the mean over test patients of the per-patient bone MAE is below 25 HU, and the gate passes if any of the six combinations does. This multiplicity is not corrected; the patient-level bootstrap 95% confidence interval is reported with each verdict, and a pass whose upper bound reaches 25 HU is reported as marginal. If several combinations pass, the one carried to Objective 3 follows an order fixed a priori rather than their errors: arcsinh encoding, then the 20000 HU ceiling, then the published windows, whose 2000 HU ceiling clips metal; within an encoding, the pretrained decoder precedes the adapted one. If none passes, the latent route is abandoned and the No-Go result is reported as the outcome. *Measured outcome.* The gate was run under this rule. On the 34 held-out patients the pretrained autoencoder reached a mean per-patient bone MAE of 160.13–209.39 HU and the decoder-adapted variant 61.72–77.03 HU; the best of the six combinations, arcsinh encoding with an adapted decoder, gave 61.72 HU (bootstrap 95% CI [55.12, 68.99]), with all 34 patients above the threshold in every combination. Identity controls reproduced the window-only reference (165/165) and the pretrained round-trip reproduced an earlier full-cohort run (330/330); encoder hashes confirm it stayed frozen and the validation curves had flattened, so the result is not an artefact of insufficient training. Inside B_δ the same adapted decoder left 69.89 HU, so restricting the latent to the generation band would not recover the loss either. The outcome is therefore **No-Go** and is reported as a result: pretrained natural-image latents do not transport dense-bone CT numbers at this tolerance. A CT-trained latent was examined next, under the same rule and on the same held-out patients, and was ruled out before running it: the released MAISI bundle (Guo et al., 2025) maps its output to $[-1000, 1000]$ HU in its own sampling script, a range the article itself does not state, so dense cortical bone above that ceiling and metal above 2500 HU fall outside the representation by construction. Clipping the held-out volumes to that range, with no model involved, already yields a mean per-patient bone MAE of 42.24 HU, above the 25 HU criterion in 11 of the 34 patients; since this is a lower bound that no choice of weights can improve, the evaluation was not run. The finding generalizes the outcome rather than being incidental to one architecture: the pretrained latents available here, including one trained on CT, are defined over intensity ranges that exclude dense bone and metal. Objective 3 responds by synthesizing in the image domain; that redesign was decided after seeing this result and is declared as such, while the gate rule itself was fixed beforehand and is not modified. This will serve as a guarantee that the intensity gradients that drive downstream applications are preserved.
- 2. Design of surgically constrained placement sampler:** propose position-orientation 3D pose of a parametric rigid screw model constrained by bone density read from the CT numbers of each volume, motivated by the systematic heterogeneity of pelvic bone mass (e.g., low grey values in the sacral ala) shown by a population-mean statistical model (Arand et al., 2019), cortical boundaries with a 5 mm clearance rule, and osseous-corridor viability (McLaren et al., 2021), expressed in the CT-computable reference frame of Kaiser et al. (2014) (sacral-axis reformat; coronal and axial corridor angulation against iliac-crest and posterior-iliac-spine landmarks). Corridor geometry is measured on anatomical masks from TotalSegmentator 2.18.0 (Wasserthal et al., 2023), which is built on nnU-Net (Isensee et al., 2021) (task `total`, default crop; robust crop as sensitivity analysis), quality-controlled per case and cleaned of connected components below 0.1% of their structure, rather than on an HU threshold. The cited publication evaluates a 104-structure model and reports no accuracy for the S1 label used here, so its reported accuracy is not assumed for the version and classes used in this work. The sampler yields a distribution of placements from a pre-registered geometric perturbation model, rather than a single artificial "optimized" trajectory; the resulting breach-grade distribution is then measured and compared against the two ordinal malposition distributions of Zwingmann et al. (2009), conditioned on surgical technique within S1. No parameter of the sampler is derived from those distributions. Sacral level remains a geometric conditioning variable, anatomically motivated by the segment-specific corridor measurements of Gardner et al. (2010), but it is treated as a measured property of each volume rather than as a fixed label. The corridor search returns, below the S1 corridor, a second osseous corridor whose vertebral level is not constant: in a depth-stratified subsample of 18 volumes from the cohort, judged on sagittal reformats by a second reader blinded to the measured depth, that second corridor fell at S2 in 13 volumes, at S3 in 4 and at S4 in 1, at median depths of 30, 40.5 and 51 mm below the S1 landmark and with median corridor diameters of 8.8, 7.4 and 3.8 mm. Depth predicts the level but does not determine it: one corridor 39 mm below S1 was judged S2 while others at 36 and 33 mm were judged S3. Results for it are therefore reported as the second corridor below S1, not as S2, and descriptively, because the clinical comparison of van den Bosch et al. (2002) concerns neurologic complaints by screw configuration and provides no ordinal placement distribution for any level below S1. This is local evidence for the same conclusion the cross-sectional areas above support: measure each corridor on the individual anatomy instead of assuming a universal level hierarchy. Sacral-morphology phenotypes are deliberately not used as a stratifying variable: their reported effect on safe-zone size is not consistent across studies, and the sampler instead measures corridor geometry directly on each volume.
- 3. Design of a conditioned synthesizer:** 2.5D conditional diffusion in the image domain, formulated as inpainting of the generation region

$G = M \cup B_\delta$, where M is the rigid implant mask. The model receives the patch with G removed, the implant mask and the band mask, and generates CT numbers inside G only; everything outside G is copied from the source volume, so preservation outside the band holds by construction rather than by measurement. This replaces the latent formulation and, with it, the ControlNet-guided Stable Diffusion 1.5 backbone: ControlNet presupposes a frozen pretrained base and its latent space (Zhang et al., 2023), Objective 1 measured that this latent does not preserve bone CT numbers at the required tolerance, and no pretrained pixel-space CT base is adopted here. The multi-window encoding is kept as the input and output channel representation, now with no compression stage between them. A crucial detail in the formulation is that the synthesis mask must be extended with a spatial band B_δ ($\sim 12\text{mm}$) over the implant boundary, to allow a more explicit manifestation of the streaking artifacts and beam hardening that occur outside of the implant mask in the generated image. The band width is a design parameter: it bounds where artifact is generated and deliberately truncates far-field streaks, whose spatial extent remains unmeasured in the reviewed literature. No published value calibrates it, but its direction is supported: in a metal-mask sensitivity study, Karageorgos et al. (2024) report that dilating the metal trace leaves reconstruction error essentially unchanged (RMSE 11.45 and 11.63 HU at dilation factors of 1.4 and 1.8), whereas shrinking it makes error collapse (54.82 and 57.69 HU at factors of 0.7 and 0.5), against 7.57 HU for the true trace. Under-covering the extent of metal is therefore far more damaging than over-covering it, which is the argument for generating beyond the implant mask rather than inside it. That experiment varies a mask in sinogram-domain artifact reduction, not a generation band in the image domain, so it motivates the direction of B_δ without fixing its width; the 12 mm value remains a declared construction. Training pairs come from the metal-bearing volumes of the training patients, where the real implant supplies both the mask and the appearance target, under the same patient-level isolation as the rest of the work; at synthesis time the mask is the parametric screw of Objective 2 placed in a sampled pose. The training unit is the individual connected metal component rather than the axial slice, so that a training region has the same structure as a synthesis region, which contains a single implant. Inclusion is governed by a criterion fixed before training: the component must lie inside the patient body, the case must carry declared orthopaedic material, and the patch must contain metal of the target component, together with a declared proportion of band-only patches. Component size and shape are reported, not filtered, because the model learns a mask-to-artifact relation that does not depend on implant type. Three domain shifts are declared rather than assumed away: training masks are obtained by thresholding real metal, whose shaft the local audit measured below the published calibres, while the synthesis mask is a smooth parametric cylinder; the context outside G carries far-field streaks in metal-bearing training volumes but not in the metal-free volumes used for synthesis; and band-only patches, which contain generation region but no metal, are less frequent in the extracted training set than the corridor geometry predicts for synthesis. The first and third are quantified and reported; the second is evaluated through the continuity of CT numbers across the band boundary.

4. **Formalization of the placement metric and adoption of an inherited appearance protocol:** Formulation of Surgical Admissibility of Placement (SAP) as the only metric introduced by this work, evaluated with Wasserstein-1 distances against the measured ordinal distributions. Appearance is evaluated with the metrics of the adopted protocol under their published names — bone integrity, metal integrity and streak amplitude (Peters et al., 2025) — and the previously proposed BFC and ISC are withdrawn, so that inherited measures are not renamed as contributions.

Explicitly out of scope. Downstream segmentation evaluation (Dice, HD95) is excluded from this study, and constraint-by-constraint ablations are deferred to future work. Two auditable conditions drive the exclusion rather than time alone: within the locally available cohort only a minority of the metal-bearing volumes carry verified bone annotation, and only part of the published collection is held locally, so a downstream result would rest on a cohort too small and too partially annotated to support the claim. The synthesis arms and the placement sampler are therefore evaluated on physical coherence and surgical admissibility alone; downstream utility is left as future work.

5. Expected Results

Pre-registration of the placement sampler. Because the placement metric is scored against clinical distributions that the sampler must not be fitted to, the sampling distribution and the metric were frozen in writing before any distance was computed, following the same procedure already used for the representation-viability gate. The sampler perturbs, rather than searches: for each volume it takes the corridor axis measured on that volume, anchors it at the midpoint of the corridor segment, and applies an isotropic perturbation in the plane perpendicular to the axis, with a half-normal magnitude truncated at three standard deviations. The whole scale derives from one independent constant, the 5 mm cortical clearance of Kaiser et al. (2014) already adopted for the reference frame, under the declared convention that this clearance corresponds to two standard deviations: a translational σ of 2.5 mm and, per case, an angular σ equal to the angle subtending that same clearance at the tip of that case's corridor, which is 2.07° at the cohort median corridor length. No parameter is taken from Zwingmann et al. (2009), deliberately including the 4° angular tolerance that its introduction quotes from a third source: it appears in the same paper against which the result is scored. Fifty poses are drawn per case, from a seed derived from the case identifier, so that every patient receives identical poses in any cohort that contains it and the primary and sensitivity cohorts differ only by their composition. The design is declared falsifiable at three points: the unperturbed pose of each case must score a breach of zero, because it is the corridor axis itself; the breach must not decrease as the cylinder is widened; and no pose may be left ungraded, since discarding the poses that miss bone would remove precisely the worst placements from the denominator and lower the reported distance for the wrong reason.

Placement result, and what the aggregate distance does and does not measure. The pre-registered analysis is the one reported in the table above, computed over the whole cohort with nothing excluded. Its integrity checks pass: 72 and 49 volumes as specified, every pose graded, and the unperturbed pose of each volume scoring a breach of zero at its own corridor diameter. The aggregate distance, however, is a mixture of two anatomically distinct populations, and reporting it alone would attribute to the sampler a mismatch it does not produce. In 15 of the 72 volumes the measured corridor is narrower than the 7.0 mm calibre at which breach is graded, so the optimal corridor axis of those patients already perforates and grade 0 is unreachable for them by the receiving anatomy rather than by placement. What narrows those corridors is not established. A blinded screen of those 15 volumes against 15 randomly drawn volumes whose corridor admits the calibre, read by a licensed physician without specialty training and with the case-to-stratum assignment withheld, found a displaced sacral fracture in 7 of the 15 narrow corridors against 2 of the 15 comparison volumes. The direction matches the mechanism Reilly et al. (2003) measured, and the effect is large, but it does not reach significance at this sample size (Fisher exact $p = 0.109$), and 5 of the 14 sacral fractures sit at S2, where that study offers nothing. The honest reading is therefore neither that the narrow corridors are normal anatomical variation nor that they are pathology: *an unquantifiable fraction of them may be displaced fracture rather than anatomy*, and this cohort cannot say how large that fraction is. Stratifying on that variable, which is a declared post-hoc analysis and not part of the pre-registration, separates the two: in the 57 volumes whose corridor admits the calibre, the sampler places 64.2 % of poses at grade 0 against 69 % for the navigated arm, a distance

Goal	Metric(s)	Expected Outcome
Surgical Admissibility (Only novel metric, executed)	SAP: four-level Cortical Breach Grade (Smith et al., 2006), with boundaries fixed by convention as grade 0 for no perforation, 1 for (0, 2), 2 for [2, 4] and 3 for (4, ∞) mm; density-zone fraction; corridor viability (McLaren et al., 2021) in the reference frame of Kaiser et al. (2014). Breach is measured at a 7.0 mm calibre as the maximum radial protrusion of the implant beyond the segmented bone envelope, over the implant’s own fixed length with 8 mm trimmed at each end, because entry and exit through the iliac cortex are by design. Placements that do not traverse bone are graded 3 rather than discarded, so that the denominator matches the clinical series, in which every screw was graded. The envelope is a bone mask, not a segmented cortex, so its boundary approximates the outer cortical surface; resolution on known geometry is 0.083 mm.	Distribution comparison: Wasserstein-1 distance between the measured breach-grade distribution of the sampled placements and each of the two ordinal distributions of Zwiggmann et al. (2009), conditioned on technique within S1. The distance is reported as measured, in grades; a large distance is a reportable outcome, because the sampler was not fitted to these distributions. Result: over the 72-volume cohort, 50 poses per volume, the sampled placements grade 51.5 / 31.4 / 11.1 / 6.0 % across grades 0–3 ($n = 3600$), at a Wasserstein-1 distance of 0.206 grades from the navigated arm and 0.230 from the conventional arm. The 49-volume sensitivity cohort gives 54.9 / 30.4 / 9.9 / 4.8 % ($n = 2450$), at 0.186 and 0.299. Every pose is graded; none is discarded. Geometry and breach grades for the second corridor below S1 are reported descriptively, without a clinical ordinal target; its vertebral level is not constant across volumes.
Appearance Coherence (Inherited protocol, published names)	Bone integrity; metal integrity; streak amplitude (Peters et al., 2025). Inversion of each metric for synthesis is documented, and the 0–4 scale is not carried over.	Streak amplitude is the single primary endpoint, declared before the test set is examined; the remaining metrics are descriptive. Superiority over naive copy-paste insertion is assessed with a one-sided paired Wilcoxon signed-rank test, and comparability with the physics-based protocol with a two one-sided test of equivalence, so that equivalence is claimed only when the paired difference lies within a margin fixed in advance, never inferred from a non-significant difference. That margin is set from the method’s own variability under conditions that should not change the result, measured on validation patients only. With the available sample sizes a non-conclusive equivalence result is a possible and reportable outcome.
Representation Viability (Compulsory gate, executed)	MAE in HU (Round-trip; bone, and B_δ descriptively)	Result: No-Go. Best of the six pre-specified combinations: 61.72 HU (95% CI [55.12, 68.99]), with 34/34 held-out patients above the threshold. The criterion was a mean per-patient bone error strictly below 25 HU, a level anchored on the CT-number accuracy reported for NMAR (20.2 HU) and for learned metal-artifact methods (12.3–12.74 HU) (Karageorgos et al., 2024; Yun et al., 2026), not on a published pass/fail criterion. Log-compressed metal ranges preserve the structural topology.
Preservation outside B_δ (No-damage check)	RMSE, SSIM against the original CT, restricted to the complement of B_δ .	No measurable change where none is intended; deviation inside B_δ is the intended signal and is not scored as error.

of 0.183 grades, and the sensitivity cohort reproduces this almost exactly at 64.6 % and 0.175 over 41 volumes; in the 15 narrow-corridor volumes grade 0 nearly vanishes, at 3.3 %, and the distance rises to 1.122. The resemblance of the aggregate figure to the conventional arm is therefore partly an artefact of that mixture, since within the viable stratum the distance to the conventional arm is 0.482. Neither figure replaces the other: the pre-registered aggregate is the result this work defends, and the stratification is reported beside it so that the anatomical component of the distance is visible. Restricting the cohort to viable corridors was considered and rejected, because selecting patients on a variable correlated with the outcome would raise the reported grade 0 fraction for the same reason that fitting the sampler to the benchmark would, and the reference series did not exclude its own narrow-corridor patients either. This finding sharpens a limitation already stated above, and the screen just described makes it concrete: the clinical distributions come from fractured pelvises, whereas the receiving anatomy here was selected as free of osteosynthesis rather than verified as free of fracture, and that screen found fracture in 20 of the 30 volumes it examined. A non-negligible fraction of the receiving anatomy therefore cannot accommodate the benchmark implant at all.

6. Datasets

Primary Dataset. CTPelvic1K (Liu et al., 2021a) will be used as the source of patient anatomy and real artifact observations. The locally available dataset6 and dataset7 volumes are inventoried separately from the full published collection; the CLINIC-metal label alone does not establish implant type or annotation availability. Acquisition and reconstruction settings, including any vendor metal artifact reduction or monoenergetic reconstruction, are not reported in the dataset publication; since both change artifact appearance (Selles et al., 2024), real artifact observations are treated as an appearance reference of unknown reconstruction, not as a physical ground truth. *Strict Isolation Rule.* No patient may contribute to both training and evaluation. The implant geometries consumed by the proposed sampler and by the physical simulation arm are parametric and are not derived from any evaluation volume. Duplicate-volume screening and patient grouping are required to enforce this separation.

Implant geometry source. Implants are modelled as parametric rigid screws rather than extracted from the metal-bearing volumes. Three geometries are kept apart, because they answer different questions. The *corridor envelope* used for placement viability is the nominal calibre reported for iliosacral fixation, 6.5–8.0 mm (Gardner et al., 2010) and 6.3–8 mm (Kaiser et al., 2014). The *metal mask* used for synthesis is the body of the implant, which is thinner: the manufacturer’s technique guide for this system prints a 4.8 mm shaft and states that core and shaft diameters coincide (Syn, 2003), a biomechanical study reports 4.9 mm shaft and 4.7 mm minor diameter under the thread for the same nominal calibre (Gardner et al., 2015), and a finite-element study of the same screw uses 7.3 mm only for the distal thread, with a 2.5 mm pitch over 16 mm, and 4.8 mm for the shaft (Zhu and Liao, 2023). The nominal figure is therefore the thread diameter, not the body, and a uniform

6.5–8.0 mm cylinder would overstate the metal cross-section along the corridor by more than a factor of two. The *measurement calibre* used to grade cortical breach is a third geometry, 7.0 mm, and it is fixed by the benchmark rather than chosen: Zwimgmann et al. (2009) graded their four levels on 7.0 mm cannulated screws. Because perforation depth scales with radius, grading simulated placements at any other calibre would fold a systematic bias into the Wasserstein-1 distance and make it uninterpretable; 4.91 and 7.3 mm are reported as sensitivity. The head is modelled from the only published measurement found, 8.0 mm in diameter and 4.5 mm high, seated without countersinking (Sayres et al., 2015). Screw length is not taken from a fixed published range: it is bounded on each volume by the measured corridor under the usable-length rule of Kaiser et al. (2014). Three parameters remain without a source in the manufacturer’s own documentation and are declared rather than assumed: washer thickness, taken as 1.5 mm from a different manufacturer’s catalogue (Dou, 2021) against an outer diameter of 13.0 mm that three sources agree on; the cannulation diameter, which is left as a free parameter of the model and evaluated solid against hollow, since the 2.9 mm value circulating in distributor listings appears in the manufacturer guide only as the cannulation of drills and screwdrivers; and the alloy, since the same system is supplied in 316L stainless steel and in Ti-6Al-7Nb (Syn, 2003) while a binary mask cannot encode which one, so the renderer learns the appearance of a mixture of materials. The real implants of the metal-bearing subset do not enter the geometry bank; they serve as a descriptive contrast for placements, as an appearance reference for synthesis, and as a test–retest control where the same implant was imaged twice. The local audit motivates this choice: screws extracted from these volumes with a fixed HU threshold appear fragmented, with shaft diameters below the published calibres, and the extracted geometry changes between acquisitions of the same implant.

Physical comparison protocol. The hybrid simulation and evaluation protocol of Peters et al. (2025), also associated with the AAPM CT-MAR Challenge (Haneda et al., 2025), is adopted instead of an independently validated reimplementation of XCIST/CatSim. Challenge data remain excluded from model training, and the scoring benchmark is reserved for final evaluation. Reproduction will document the code revision, configuration, reconstruction settings and access conditions. The physical arm is run on a reduced evaluation subset of patients. Both synthesis arms will use the same source anatomy and implant poses; a no-metal simulation/reconstruction control will quantify changes introduced by the physical arm alone. The published protocol is two-dimensional and evaluates MAR: extending it to pelvic osteosynthesis synthesis requires an explicit adaptation and does not inherit its validation automatically. RMSE and SSIM against the original CT will assess preservation outside the generation band, where changes are not intended; artifact realism within the affected region requires separate evaluation. No unverified numerical acceptance thresholds or MAR score calibration are adopted here.

Field limitation: corridor geometry has almost only been characterized in intact pelvises. This is not a limitation of the present cohort but of the available literature. With one exception, every source that publishes sacral corridor geometry measured it in pelvises without implants and without fracture: Kaiser et al. (2014) excluded scans with implants obscuring the lumbosacral junction, McLaren et al. (2021) used uninjured scans, and Ziran et al. (2007) worked on cadaveric specimens and warn that fracture, displacement and soft-tissue injury would alter what is visible. The exception, Reilly et al. (2003), osteotomised a zone II sacral fracture in six cadaveric pelvises and found that 5–20 mm of cranial displacement reduced the cross-sectional area available to S1 iliosacral screws by 36–90%, so corridor limits obtained on intact pelvises do not transfer to fractured or malreduced ones; this supports measuring the corridor on each volume rather than importing a published limit. It also exposes an uncontrolled difference between benchmark and host: the clinical ordinal distributions of Zwimgmann et al. (2009) come from fractured pelvises (Tile types B and C), whereas corridors here are measured on pelvises without osteosynthesis, so any residual malreduction in the clinical cohort narrows its corridors in a way the synthetic hosts do not reproduce. The reference frame adopted here rests on landmarks at the lumbosacral junction, which is precisely where streaking concentrates, so whether those landmarks remain locatable in metal-bearing volumes is not answered by any published source. It was therefore quantified on the local cohort. Landmarks were located by a heuristic on each volume, and a landmark region was counted as contaminated when its dark-streak or metal fraction exceeded the maximum observed in 69 metal-free patients of the same cohort. Among 65 patients with orthopaedic metal, all five landmarks were found within the field of view in 57. After a clinician reviewer (one surgeon, otorhinolaryngology), blinded to the automatic audit, confirmed the S1 level on every patient, the frame was computable in 48 patients and free of contamination at all five landmark regions in 29. A second, independent read of the same 61 landmarks on sagittal reformats agreed with that reviewer in 61 of 61 cases; the second reader was the author rather than a second clinician, and only the level judgements are independent, since the accompanying free-text notes were copied across once the judgements were seen to coincide. Losses were driven more by field-of-view truncation of the iliac crests (7 patients) and by landmark localisation than by artifact contamination; in the 17 patients with a screening-detected iliosacral screw, the S1 level was correct in all and its region was contaminated in 3. The frame is therefore computed on metal-bearing volumes where it is computable. The field-of-view-truncated cases require no separate handling for placement: of the volumes on which corridors were measured, those flagged for iliac-crest truncation all fall outside the cohort used for the placement sampler, so they enter none of its results. Finally, the heuristic localisation of the iliac crests and posterior iliac spines has not been clinically reviewed.

Corridor measurement on the local cohort. Among 103 patients without osteosynthesis, 91 had a located S1 and 72 passed level quality control. Level discordance excluded 11 of 34 patients with non-orthopaedic objects and 7 of 57 clean patients, and one further patient was excluded because S1 reached the field-of-view border; this imbalance is declared. The maximum transsacral corridor diameter had a median of 9.5 mm (IQR 7.4–11.7), and the 10 mm convention was met in 29 of 72 patients with the default crop and in 27 with the robust crop; the crop is therefore reported as a measurement uncertainty that reaches the binary viability outcome. Component cleaning changed neither the transsacral corridor diameter nor its 10 mm viability in this cohort at any tested fraction up to 5%; the fraction, like the occupancy rule below, was fixed after inspecting these results. Along the axis of the widest corridor, the median fraction of voxels at or below 150 HU inside the sacral masks was 0.42, so an HU-based bone definition would exclude much of the corridor. In metal-bearing volumes, where the segmentation labels the existing implant as bone, treating the implant as occupied space (local half-maximum threshold per object) lowered 10 mm viability from 53.8% to 40.4% in 52 patients with the default crop, and a fixed 2500 HU threshold, which thins the screws and therefore bounds the occupied corridor from above, gave the same proportion; these volumes are reported as a separate contrast and are not pooled with the corridor cohort.

Local cohort audit. Training eligibility requires manual review of three-dimensional views and multiplanar CT slices confirming absence of metal and foreign objects. HU screening only prioritizes this review, including cases from dataset6. Confirmed metal cases are candidates for testing, with pelvic osteosynthesis distinguished from other implants and external objects. Exact volume duplicates and repeated patients must be resolved before splitting. Final cohort sizes remain pending this audit.

Why downstream evaluation is excluded. The audit itself supplies the reason. Only a minority of the metal-bearing volumes carry verified bone annotation, and only a fraction of the published collection is available locally, so a Dice or HD95 result would rest on a cohort too small and too partially annotated to support a robustness claim. The metal-bearing subset used here is nonetheless the de facto clinical evaluation set

for pelvic metal-artifact work, having been used as such in recent published work (Wang et al., 2025). Where a clean reference image does not exist for these volumes, appearance realism is assessed against real artifact observations rather than against an unavailable ground truth.

7. References

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